PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	COMMUNICATION IN DECISION AIDS FOR STAGE I-III COLORECTAL CANCER PATIENTS: A SYSTEMATIC REVIEW
AUTHORS	Hommes, Saar; Vromans, Ruben; Clouth, Felix; Verbeek, Xander; de Hingh, Ignace; Krahmer, Emiel

VERSION 1 – REVIEW

REVIEWER	Ciarán D. McInerney University of Leeds, UK
REVIEW RETURNED	09-Oct-2020

GENERAL COMMENTS	Page 5, Line 57: Typo - “Das” should be “DAs”. Page 8, Line 51: Please prefix “academic” to “articles” to make it clear that this initial discussion is with respect to the sources obtain via the literature search of academic databases. Page 8, Line 58: Typo – “inter-rate” should be “inter-rater”. Page 10, Line 24: Typo – “inter-rate” should be “inter-rater”. Page 10, Line 26: It is not clear to whom the reader should attribute the Kappa statistics provided. Please, provide a table (or succinct prose) exactly detailing the agreement indices for each subgroup. What are the subgroups, exactly? (IPDAS + CA) x (2 coders) x (4 groups)? Page 10, Line 33: Regarding citation 20, rather than saying he IPDAS is validated, please provide the statistics reported in the assessment of its validity so that the reader can decide for themselves whether or not the IPDAS is sufficient. Page 10, Line 37: Please, replace “not all DAs were academically validated” with “the validity of DAs was not assessed academically for all DAs”. The current language used suggests that tools can be bestowed with validity, which is not true; instead, the validity of tools is assessed. Page 10, Line 37: I suggest it is better to include the IPDAS-Evaluation dimension and report it as NULL for those DAs that did not have their validity assessed. The lack of evaluation is clearly important enough for the authors to mention in the text so it should be important enough to incorporate into this study’s protocol and interpretation. If NULL values make interpretation of IPDAS difficult, please reflect and report on such short-comings of the tool (in my experience, many quality-appraisal tools have faults in their closed questions and constrained set of permitted responses). Page 24, Line 5: Please, don’t use “M” and “SD” to indicate mean and standard deviation, especially because there are already a lot of initialisms being used in this article, with this paragraph exemplifying why it can make communication more difficult. I would suggest that it is more informative to report median, 1st quartile and 3rd quartile to give the reader a more insightful understanding

	of the distribution of scores. Reporting mean and standard deviation is only relevant if the distribution is Gaussian (Normal) Page 25, Figure 2 & Page 27 Figure 3: Please, remove the data points because they are a bit messy and the count of data is sufficiently communicated by the width of the violin plots. Please, also include provide a label for the vertical axis. Please, use add theme_classic() to the R script being used to plot the figure because it will remove the unnecessary grey plot background. Finally, remove the plot title because the figure has a caption (I appreciate that the plot title was useful during initial working but the figure caption takes the title's role in publication). Page 30, Line 54: I don't think the authors are sufficiently justified to conclude that the IPDAS and CA checklist are useful tools for guide developers. At the very least, please add the implied rationale that, as measurement tools, they can guide design requirements. 1. Well-structured and clearly presented work. 2. Please, can you include the authors working definition of what 'communicative quality' is? 3. No explicit rationale for focusing on stage 1-3. Can the authors please provide one? What are stage 4 decision aids qualitatively different? 4. Regarding inter-rater agreement, please provide the contingency table/agreement matrix that informed the calculation of the Kappa statistic so that readers can compute their own agreement indices. I would encourage the authors not to rely on the Kappa statistic because its assumptions are untenable (Feinstien & Cicchetti, 1990). Superior and more-informative alternatives would be raw agreement indices to test trait agreement (Cicchetti & Feinstein, 1990), and the Stuart-Maxwell test for marginal homogeneity to test for category agreement (Sun & Yang, 2008). Log-linear analysis and comparing odds ratios are also good options provided that no cell in the contingency tables are zero. References are below. Cicchetti, D. V., & Feinstein, A. R. (1990). High agreement but low Kappa: II. Resolving the paradoxes. Journal of Clinical Epidemiology, 43(6), 551–558. Feinstien, A., & Cicchetti, D. V. (1990). High Agreement but Low Kappa: I. The problems of two paradoxes. Journal of Clinical Epidemiology, 43(6), 543–549. Sun, X., & Yang, Z. (2008). Generalized McNemar's Test for Homogeneity of Marginal Distributions. Statistics and Data Analysis, SAS Global Forum 2008, Vol. 382. Retrieved from http://www2.sas.com/proceedings/forum2008/382-2008.pdf 5. Regarding discussion on the formality of languages, could the authors please review and summarise relevant literature on the cultural differences of how authority figures like doctors are perceived? This seems very relevant and worth much more than a passing comment, especially if the authors wish to evaluate the global generalisability of the PROFILES tool. 6. The initialism "SDM" is used so infrequently that it hinders understanding. Please, write 'shared decision making' in full for all of the only four uses of the term. 7. The authors might find it useful to review the literature on design principles for decision –support tools/systems, especially as they progress the evaluation of PROFILES - although I appreciate that decision aids are a subset. Some suggested reading is given below. For example, Bates et al. (2003) provide ten

	commandments for effective clinical decision support, most of which relate to the interaction between the user and the software, which take a distinct user-centred design (e.g. Griffin et al., 2019). Such user-centred design is mandated for electronic health record systems by the Office of the National Coordinator of Health Information Technology, part of the U.S. Department of Health and Human Services (United States Department of Health and Human Services, 2013). Horsky et al. (2012) provide design principles for clinical prescribing decision-support software but they are applicable to all interface designs. In their review of models and frameworks for clinical decision support, Greenes et al. (2018) distinguish argumentation-logic versus probabilistic designs for decision support. The former provides pros and cons associated with the clinical decision of interest, while the latter provides a measure of the uncertainty of the decision-support software's estimate. Given patient's and clinician's difficulty with and differences in interpreting probability, risk and risk scores, the argumentation-logic design might be better for engaging users, at the cost of not being the standard type of output from statistical models. It might be better to front-load the difficult work of developing argumentation-logic, decision-support software, rather than produce probabilistic, decision-support software with relative ease but with the accompanying burden on the user. van de Velde et al. (2018) provide a 16-item checklist for implementing decision-support software that was informed by a literature review by an expert panel, and considers the context, content, features and delivery of the software. Bates, D. W. et al. (2003) 'Ten Commandments for Effective Clinical Decision Support: Making the Practice of Evidence-based Medicine a Reality', Journal of the American Informatics Association, 10(6), pp. 523–530. doi: 10.1197/jamia.M1370.Although. Greenes, R. A. et al. (2018) 'Clinical decision support models and frameworks: Seeking to address research issues underlying implementation successes and failures', Journal of Biomedical Informatics, 78(July 2017), pp. 134–143. doi: 10.1016/j.jbi.2017.12.005. Griffin, L. et al. (2019) 'Creating an mHealth App for Colorectal Cancer Screening: User-Centered Design Approach', JMIR Human Factors, 6(2), p. e12700. doi: 10.2196/12700. Horsky, J. et al. (2012) 'Interface design principles for usable decision support: A targeted review of best practices for clinical prescribing interventions', Journal of Biomedical Informatics. Elsevier Inc., 45(6), pp. 1202–1216. doi: 10.1016/j.jbi.2012.09.002. United States Department of Health and Human Services (2013) 'HIT Standards Committee and HIT Policy Committee Implementation and Usability Hearing'. Washington, DC: Department of Health and Human Services - Office of the National Coordinator for Health Information Technology, pp. 1–87. van de Velde, S. et al. (2018) 'The GUIDES checklist: development of a tool to improve the successful use of guideline-based computerised clinical decision support', Implementation Science. Implementation Science, 13(86). 8. Please, include the completed PRISMA checklist. 9. Please, discuss further the limitation of using the IPDAS and CA checklist to infer communicative quality.

REVIEWER	Retnagowri Rajandram UNIVERSITY OF Malaya
REVIEW RETURNED	03-Nov-2020

GENERAL COMMENTS	Dear Authors, Thank you for the ability to be one of the reviewer for your systematic review titled "COMMUNICATION IN DECISION AIDS FOR STAGE I-III COLORECTAL CANCER PATIENTS: A SYSTEMATIC REVIEW" This was an interesting read and clearly highlights the gap in knowledge that would need further work in the near future. It also provided some insight into the usefulness and the value of the Decision Aid in relation to Colorectal Cancer Stage I-III. The content itself is sound. However there are some noticeable typing errors in the manuscripts. Also the keywords in the abstract section could be re-arranged in alphabetical order. Please do address these minor recommendations.

VERSION 1 – AUTHOR RESPONSE

Reviewer 1:

We would like to thank reviewer 1 for the thorough and fair feedback, we really appreciate the time spent reading our manuscript and we think our manuscript greatly benefitted from all of these suggestions. We have incorporated almost all of the suggestions (if we have not incorporated a suggestion, we hope you are satisfied with the reasoning behind this, described below).

1. **Page 5, Line 57: Typo - "Das" should be "DAs".**

We have removed this typo.

2. **Page 8., Line 51: Please prefix "academic" to "articles" to make it clear that this initial discussion is with respect to the sources obtain via the literature search of academic databases.**

That indeed makes sense. We changed the sentence to:

"For both the academic articles and the online search, only tools aimed at supporting the decision-making process of colon, rectal and colorectal cancer patients stage I - III were eligible for selection."

3. **Page 8, Line 58: Type – "inter-rate" should be inter-rater".**
4. **Page 10, Line 24: Typo – "inter-rate" should be "inter-rater".**

Thank you for spotting these typos, we changed them both.

5. **Page 10, Line 26: It is not clear to whom the reader should attribute the Kappa statistics provided. Please, provide a table (or succinct prose) exactly detailing the agreement indices for each subgroup. What are the subgroups, exactly? (IPDAS + CA) x (2 coders) x (4 groups)?**

We have now included a table that outlines which coding-team had which Kappa in Appendix 4 (see also below):

Review Team	***DAs***	***Kappa IPDAS***	***Kappa CA***
SH – RV	*N = 7 (DA: 1; 8; 9; 13; 14; 16; 17)*	*0.6 (0.597)*	*0.8 (0.766)*
FC – MB	*N = 6 (DA: 2; 3; 4; 5; 6; 7)*	*0.4 (0.388)*	*0.8 (0.793)*
SH – EK	*N = 2 (DA: 10; 15)*	*0.5 (0.49)*	*0.6 (0.585)*
SH - XV	*N = 2 (DA: 11; 12)*	*0.3 (0.315)*	*0.4 (0.38)*
SH - FC	*N = 1 (DA: 18)*	*0.4 (0.4)*	*0.6 (0.57)*

6. **Page 10, Line 33: Regarding citation 20, rather than saying he IPDAS is validated, please provide the statistics reported in the assessment of its validity so that the reader can decide for themselves whether or not the IPDAS is sufficient.**

We have now included reference 19 in addition to reference 20, so interested readers can look at the validation process in more detail (since there have been a lot of validation studies done, it's not that easy to provide statistical evidence without spending two paragraphs on the process). The paragraph now reads:

"The IPDAS instrument consists of 36 items (see Table 1). It was developed by a group of clinical researchers, practitioners and stakeholders[18] to ensure that DAs adhere to certain quality standards[19] and has been validated (for more detailed information on the validation process see: Elwyn et al., 2009 and the associated website http://ipdas.ohri.ca)[19-20]."

7. **Page 10, Line 37: Please, replace "not all DAs were academically validated" with "the validity of DAs was not assessed academically for all DAs". The current language used suggests that tools can be bestowed with validity, which is not true; instead, the validity of tools is assessed.**

That is indeed better, we changed the sentence to:

"As the validity of DAs was not assessed academically for all DAs, the evaluation dimension was excluded from analysis"

8. **Page 10, Line 37: I suggest it is better to include the IPDAS-Evaluation dimension and report it as NULL for those DAs that did not have their validity assessed. The lack of evaluation is clearly important enough for the authors to mention in the text so it should be important enough to incorporate into this study's protocol and interpretation. If NULL values make interpretation of IPDAS difficult, please reflect and report on such short-comings of the tool (in my experience, many quality-appraisal tools have faults in their closed questions and constrained set of permitted responses).**

This would indeed be an option. However, we did not include the evaluation dimension because this dimension *requires* that there is a paper (the IPDAS items are: "There is evidence that the patient decision aid improves the match between the preferences of the informed patient and the option that is chosen" and "There is evidence that the patient decision aid helps patients improve their knowledge about options' features"). We felt that it would therefore entail that we had to adjust our search strategy so that all DAs at least have the same potential of scoring the maximum points. Additionally, the IPDAS creators seem to view their instrument as rather dynamic, as there are different versions for different settings (e.g., there is also a dimension for "Test", which zooms in on test results, but that assumes that the DA is used before diagnosis so that dimension was also not used for this assessment). Finally, we think it is important that all three systematic reviews that use the same assessment instruments

(Vromans et al 2019a; 2019b and the current review) use the same scoring system so that results are comparable (and we can see, for example, that DAs in CRC overall seem to score lower on all dimensions than DAs in breast cancer and prostate cancer). In Figure 1 below it is visible that the only change would be that an extra dimension is visible in the IPDAS assessment.

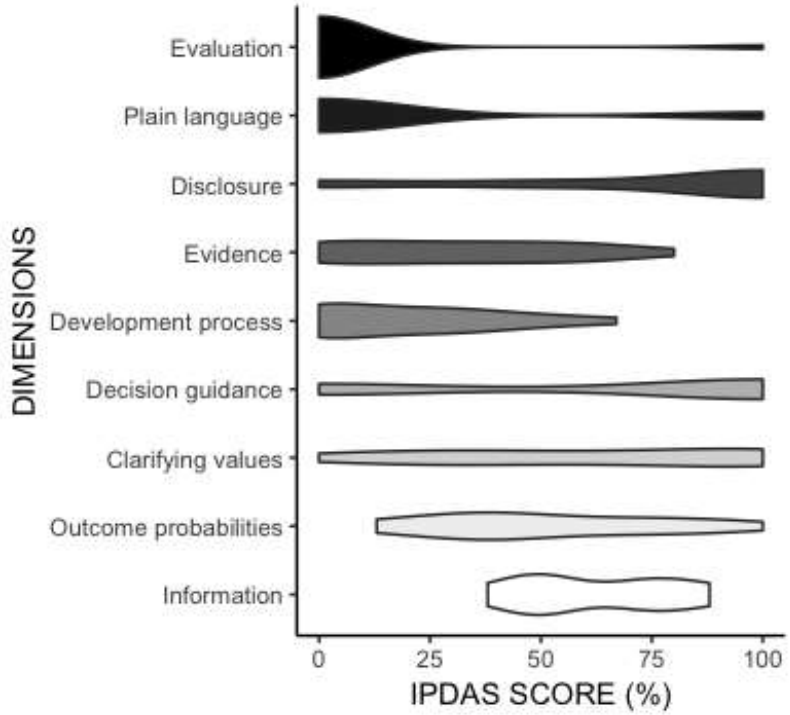


Figure 1: IPDAS results if the evaluation dimension was added to the assessment

9. **Page 24, Line 5: Please, don't use "M" and "SD" to indicate mean and standard deviation, especially because there are already a lot of initialisms being used in this article, with this paragraph exemplifying why it can make communication more difficult. I would suggest that it is more informative to report median, 1st quartile and 3rd quartile to give the reader a more insightful understanding of the distribution of scores. Reporting mean and standard deviation is only relevant if the distribution is Gaussian (Normal)**

Thank you for pointing that out, we agree. We have now written "M" and "SD" in full and included Q1, Q3 and the median. We decided to leave in reporting of mean and standard deviation to make the comparison between our results and those of the previous systematic reviews using both IPDAS and CA (Vromans et al 2019a; 2019b) better comparable. These are the adjusted paragraphs:

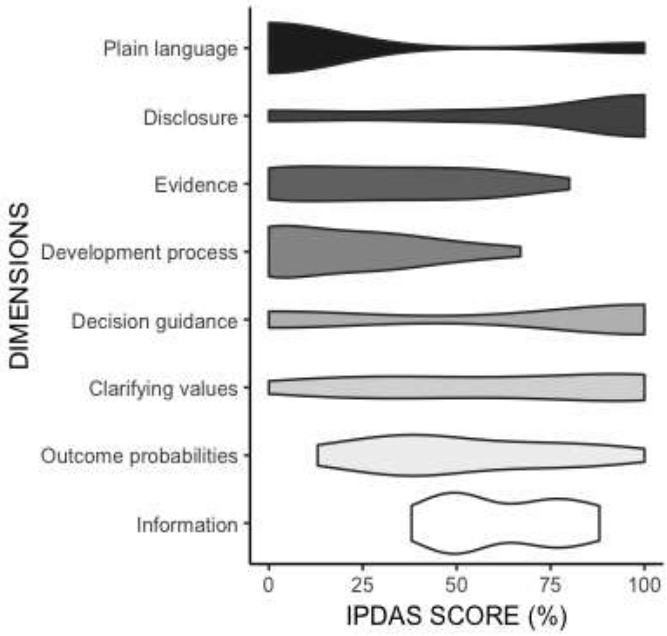
"As is visible from Table 4, IPDAS scores for individual DAs ranged from 28% to 78% (mean = 48%, standard deviation = 14.15%, 1st quartile = 38%, 3rd quartile = 58%, median = 47%)."

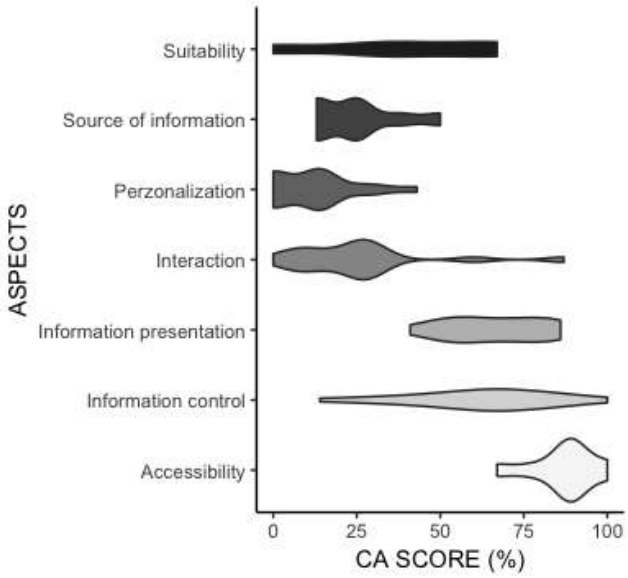
And:

"Results for the CA checklist ranged from 28% to 58% (mean = 41%, standard deviation = 6.2%, 1st quartile = 38%, 3rd quartile = 43%, median = 41%)."

10. **Page 25, Figure 2 & Page 27 Figure 3: Please, remove the data points because they are a bit messy and the count of data is sufficiently communicated by the width of the violin plots. Please, also include provide a label for the vertical axis. Please, use add theme_classic() to the R script being used to plot the figure because it will remove the unnecessary grey plot background. Finally, remove the plot title because the figure has a caption (I appreciate that the plot title was useful during initial working but the figure caption takes the title's role in publication).**

Thank you for these tips, we have now changed the figures accordingly to:





11. **Page 30, Line 54: I don't think the authors are sufficiently justified to conclude that the IPDAS and CA checklist are useful tools for guide developers. At the very least, please add the implied rationale that, as measurement tools, they can guide design requirements.**

We have now expanded on this as follows:

"To ensure this, both the IPDAS instrument and CA checklist can be useful tools to guide DA developers in such a way that they are made aware of certain aspects and can take them into account."

12. **Please, can you include the authors working definition of what 'communicative quality' is?**

We have added the following explanation to explain what communicative quality means in the CA checklist:

"With 'communicative quality' they mean the assessment of whether or not there is evidence that "the communicative process in which shared decision-making occurs (Vromans et al., 2019a, p.2)" is sufficient."

13. **No explicit rationale for focusing on stage 1-3. Can the authors please provide one? What are stage 4 decision aids qualitatively different?**

That is a very good question and one we did not answer in the original manuscript. Stage I-III CRC decision aids are distinctly different from stage IV DAs because different decisions need to be made. In stage I-III, the treatments have curative intent (e.g. patients can be cured after the treatment), whereas this is not (always) the case in stage IV. Here, the cancer has spread and the disease is metastatic. Decisions can be focused on cure (in some cases), but also on improving quality of life or managing the final stages of life. This impacts the DA because (1) the content is very different and (2) the communication is different. One might expect less statistics for example, since time periods are shorter (e.g., patients might only have a few weeks or months left). That is why screening DAs were also not taken into account, since the decision here is to undergo screening or not so people have not been diagnosed yet.

We have now included the following short explanation in the introduction:

"The focus of the current review is on stage I-III CRC, where curative treatment is the main goal (and are therefore distinctly different from stage IV DAs)."

We expand on this in the Method section:

"These tools are focused on inherently different decisions (e.g., "should I get a screening test" or "which treatments can help with quality of life for my final stages of life") and therefore require different communication strategies."

14. **Regarding inter-rater agreement, please provide the contingency table/agreement matrix that informed the calculation of the Kappa statistic so that readers can compute their own agreement indices. I would encourage the authors not to rely on the Kappa statistic because its assumptions are untenable (Feinstien & Cicchetti, 1990). Superior and more-informative alternatives would be raw agreement indices to test trait agreement (Cicchetti & Feinstein, 1990), and the Stuart-Maxwell test for marginal homogeneity to test for category agreement (Sun & Yang, 2008). Log-linear analysis and comparing odds ratios are also good options provided that no cell in the contingency tables are zero. References are below.**

Cicchetti, D. V., & Feinstein, A. R. (1990). High agreement but low Kappa: II. Resolving the paradoxes. Journal of Clinical Epidemiology, 43(6), 551–558.
Feinstien, A., & Cicchetti, D. V. (1990). High Agreement but Low Kappa: I. The problems of two paradoxes. Journal of Clinical Epidemiology, 43(6), 543–549.
Sun, X., & Yang, Z. (2008). Generalized McNemar's Test for Homogeneity of Marginal Distributions. Statistics and Data Analysis, SAS Global Forum 2008, Vol. 382. Retrieved from http://www2.sas.com/proceedings/forum2008/382-2008.pdf

Thank you for all of these options, we truly appreciate it. Even though we understand the comment from the reviewer, we prefer to keep the kappa statistics as well since they are well-understood and frequently used to compute agreement. We have, however, also provided the agreement matrices in the supplementary data (appendix V), so that readers can calculate their own agreement indices that they seem fit. As an example, the agreement matrices for Team 1 are copied here:

IPDAS Agreement Matrices (dimensions were scored with 0 = "no", 1 = "yes")

DA1:

SH = (1,1,1,0,1,1,1,1,1,1,1,1,1,0,1,1,1,0,0,1,1,1,1,0,1,0,1,0,1,0,0,0,0,1,1,1)

RV = (1,1,1,0,1,1,1,1,1,1,1,0,1,1,1,1,1,0,0,1,1,0,1,0,1,1,1,0,1,1,0,0,1,1,1,1)

DA8:

SH = (1,1,1,1,1,1,1,0,1,0,0,0,0,1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0)

RV = (0,1,1,1,1,1,1,0,1,0,0,0,0,1,1,0,1,0,0,0,0,0,0,0,0,0,0,0,0,0,1,0,0,0,0,0)

[…]

CA Agreement Matrices (aspects were scored with 0 = "no", 1 = "yes")

DA1:

SH = (1,0,1,1,0,0,0,0,1,0,1,0,1,0,1,0,0,0,0,1,1,0,0,0,1,0,1,0,0,1,1,1,0,0,1,1,0,0,0,0)

RV = (1,1,1,1,0,0,1,0,1,0,1,0,1,0,0,0,0,0,0,1,1,0,1,0,1,0,1,0,0,1,1,1,0,0,1,1,0,0,0,0)

DA8:

SH = (1,1,0,1,0,0,0,1,1,0,1,0,1,0,1,0,0,0,0,0,1,0,0,0,1,0,0,1,0,1,1,1,1,1,0,1,0,0,0,0,1)

RV = (1,1,0,1,0,0,0,1,1,0,1,0,1,0,1,0,0,0,0,0,0,0,1,0,1,0,1,0,0,1,0,1,1,1,1,1,0,1,0,0,0,0,0)

[…]

Additionally, the table provided in the supplementary materials is also copied here:

Review teams	*N of DAs reviewed*	*IPDAS disagreements per DA*	*Total CA disagreements*
Team 1: SH & RV	*7*	*6; 4; 5; 10; 8; 9; 6 (total=48)*	*4; 3; 5; 3; 6; 4; 7 (total=32)*
Team 2: FC & MB	*6*	*7; 14; 10; 11; 10; 8 (total=60)*	*4; 4; 3; 6; 2; 4 (total=23)*
Team 3: SH & EK	*2*	*11; 5 (total=16)*	*8; 7 (total=15)*
Team 4: SH & XV	*2*	*7; 15 (total=22)*	*13; 11 (total=24)*
Team 5: SH & FC	*1*	*11*	*8*

Table 1*: Overview of disagreement per review team. The labels are: review teams consisting of two reviewers (SH; RV; FC; MB; XV; EK); total number of DAs each team has reviewed; team disagreements IPDAS results per individual DA (with total disagreement of all DAs reviewed between brackets); team disagreements CA results per individual DA (with total disagreement of all DAs reviewed between brackets)*

15. **Regarding discussion on the formality of languages, could the authors please review and summarise relevant literature on the cultural differences of how authority figures like doctors are perceived? This seems very relevant and worth much more than a passing comment, especially if the authors wish to evaluate the global generalisability of the PROFILES tool.**

We have included the following paragraph in the Discussion:

"Although it has been demonstrated that culture might impact the effectiveness of health communication between doctors and patients of different cultural background[37-38], studying cultural differences between European countries remains challenging as theories and methods for assessing differences vary between countries[39]. Future reviews could look into systematic differences between DAs from different countries more to see if shared decision-making is a globally agreed upon goal."

16. **The initialism "SDM" is used so infrequently that it hinders understanding. Please, write 'shared decision making' in full for all of the only four uses of the term.**

We have changed all uses of ‘SDM’ to ‘shared decision-making’.

17. **The authors might find it useful to review the literature on design principles for decision –support tools/systems, especially as they progress the evaluation of PROFILES - although I appreciate that decision aids are a subset. Some suggested reading is given below. For example, Bates et al. (2003) provide ten commandments for effective clinical decision support, most of which relate to the interaction between the user and the software, which take a distinct user-centred design (e.g. Griffin et al., 2019). Such user-centred design is mandated for electronic health record systems by the Office of the National Coordinator of Health Information Technology, part of the U.S. Department of Health and Human Services (United States Department of Health and Human Services, 2013).**
Horsky et al. (2012) provide design principles for clinical prescribing decision-support software but they are applicable to all interface designs.
In their review of models and frameworks for clinical decision support, Greenes et al. (2018) distinguish argumentation-logic versus probabilistic designs for decision support. The former provides pros and cons associated with the clinical decision of interest, while the latter provides a measure of the uncertainty of the decision-support software’s estimate. Given patient’s and clinician’s difficulty with and differences in interpreting probability, risk and risk scores, the argumentation-logic design might be better for engaging users, at the cost of not being the standard type of output from statistical models. It might be better to front-load the difficult work of developing argumentation-logic, decision-support software, rather than produce probabilistic, decision-support software with relative ease but with the accompanying burden on the user.
van de Velde et al. (2018) provide a 16-item checklist for implementing decision-support software that was informed by a literature review by an expert panel, and considers the context, content, features and delivery of the software.
Bates, D. W. et al. (2003) ‘Ten Commandments for Effective Clinical Decision Support: Making the Practice of Evidence-based Medicine a Reality’, Journal of the American Informatics Association, 10(6), pp. 523–530. doi: 10.1197/jamia.M1370.Although.
Greenes, R. A. et al. (2018) ‘Clinical decision support models and frameworks: Seeking to address research issues underlying implementation successes and failures’, Journal of Biomedical Informatics, 78(July 2017), pp. 134–143. doi: 10.1016/j.jbi.2017.12.005.
Griffin, L. et al. (2019) ‘Creating an mHealth App for Colorectal Cancer Screening: User-Centered Design Approach’, JMIR Human Factors, 6(2), p. e12700. doi: 10.2196/12700.
Horsky, J. et al. (2012) ‘Interface design principles for usable decision support: A targeted review of best practices for clinical prescribing interventions’, Journal of

Biomedical Informatics. Elsevier Inc., 45(6), pp. 1202–1216. doi: 10.1016/j.jbi.2012.09.002.
United States Department of Health and Human Services (2013) 'HIT Standards Committee and HIT Policy Committee Implementation and Usability Hearing'. Washington, DC: Department of Health and Human Services - Office of the National Coordinator for Health Information Technology, pp. 1–87.
van de Velde, S. et al. (2018) 'The GUIDES checklist: development of a tool to improve the successful use of guideline-based computerised clinical decision support', Implementation Science. Implementation Science, 13(86).

Thank you very much for these reading suggestions, we will definitely incorporate them in the evaluation of the tool we are working on. We are especially interested in the argumentation-logic because we have also encountered a subgroup of patients that simply does not seem to be interested in (personalized) statistics, and we are looking for ways to contextualize the statistics in such a way that more patients might benefit from the information. Additionally, we added Horsky et al (2012) as a reference since we feel their evaluation is most applicable to patient-users as well as clinicians (see comment #19 for how the reference is used).

18. **Please, include the completed PRISMA checklist.**

We have included the completed PRISMA checklist.

19. **Please, discuss further the limitation of using the IPDAS and CA checklist to infer communicative quality.**

We have added the following paragraph:

"It should also be noted that although IPDAs and CA can be used to guide the design process of DA developers, using these tools does not ensure (communicative) quality. We therefore stress that DAs should also always be evaluated with clinical experts and patients[36]."

Reviewer 2:

Thank you for the ability to be one of the reviewer for your systematic review titled "COMMUNICATION IN DECISION AIDS FOR STAGE I-III COLORECTAL CANCER PATIENTS: A

SYSTEMATIC REVIEW"

1. **This was an interesting read and clearly highlights the gap in knowledge that would need further work in the near future. It also provided some insight into the usefulness and the value of the Decision Aid in relation to Colorectal Cancer Stage I-III. The content itself is sound.**

Thank you for these kind words!

2. **However there are some noticeable typing errors in the manuscripts.**

We hope we have spotted all typing errors for this version of the manuscript.

3. **Also the keywords in the abstract section could be re-arranged in alphabetical order. Please do address these minor recommendations.**

We have re-arranged the key-words in alphabetical order:

"Key words: colon cancer; decision support techniques; health communication; personalized health; rectal cancer; shared decision-making."

VERSION 2 – REVIEW

REVIEWER	McInerney, Ciarán University of Leeds
REVIEW RETURNED	09-Mar-2021

GENERAL COMMENTS	The authors have provided an excellent and thorough response, and done very well to incorporate reviewers' feedback within the word limit. While I still might argue for some different choices, I will save most of my comments for a collegiate discussion, someday. I wish the authors all the best with their continued investigations. I have attached some parting comments that I leave with the authors to consider.

REVIEWER	Rajandram, Retnagowri University of Malaya, Department of Surgery
REVIEW RETURNED	11-Mar-2021

GENERAL COMMENTS	Thank you for the giving me the opportunity to read this manuscript. It was well written. There are some noticeable typing

	errors (e.g. the spelling of the word "Strengts" on page 2 line 50 and page 4 line 4 the word " clinical pracitce." to point out a couple of errors, please do check the document full for similar errors. The final papers used were 18, any reason the region does not include Asian countries that publish in English or even any paper from England?

VERSION 2 – AUTHOR RESPONSE

Reviewer 1:

1. **The authors have provided an excellent and thorough response, and done very well to incorporate reviewers' feedback within the word limit. While I still might argue for some different choices, I will save most of my comments for a collegiate discussion, someday. I wish the authors all the best with their continued investigations. I have attached some parting comments that I leave with the authors to consider**

We really appreciate your constructive feedback and look forward to the collegiate discussion in the future. This is how we addressed the other comments:

2. **Page 2, Line 98: "…that focusses on such treatment CRC DAs" might read better as "…that focusses on CRC DAs for treatment". Have I understood your intended meaning, correctly?**

That is indeed correct, we have changed it to:

"Currently, we are aware of only one systematic review that focusses on CRC DAs for treatment."

3. **Page 9, Line 160 - 162: The authors provided a justification for their use of the Kappa statistic in their response to reviewers' comments, which motivated the authors to submit Appendix IV and V. Given that the authors are now aware that the agreement index they have chosen is flawed, it is incumbent upon them not to ignore this information and not to promote continued use without acknowledging the limitations. I suggest that the authors use the provided references to explicitly note the short-**

comings of the Kappa statistic, but then to provide their reasonable argument for why they continue to use it, i.e. "we prefer to keep the kappa statistics…since they are well-understood and frequently used to compute agreement". Such an approach would not require additional statistical analysis but would benefit the reader's education and demonstrate the authors' commitment to upload academic integrity.

We see your point and have now included the following paragraph in the Method section:

"To calculate inter-rater agreement between team members, we used the Kappa statistic (κ). Although there has been some debate about the assumptions underlying the Kappa statistic[19-20], we decided to keep the measure as they are well-understood and frequently used to compute inter-rater agreement. We have, however, also provided the agreement matrices so other agreement indices may be calculated (see Supplementary File Appendix 5)."

4. **Page 9, Line 163: It is my experience that the resulting agreement indices should not be presented within the Methods section of the paper; they should be presented in the Results section. This might be an editorial preference but please check author guidance.**

Upon checking other articles in BMJ Open we indeed found kappa statistics are most often reported within the Results section, we have therefore removed them from the Method into the Results section.

"1. The IPDAS results

Inter-rater agreements (κ) between teams ranged from fair to substantial agreement (κ = .32 to κ = .60) for IPDAS."

"2. The CA results

Inter-rater agreement (κ) for the CA checklist ranged from fair to substantial (κ = .38 to κ = .79)."

5. **Page 18, Line 202: I admit ignorance but please check with the editor whether it is appropriate to include information about the updated search using an asterisk and footnote. I don't see a problem with including the explanatory paragraph within the main text.**

We initially thought it might be confusing for readers, but we have now decided to put the paragraph in the main text (under Results):

*"Additionally**,** we updated the search between submission and revision. On February the 16th 2021, we ran the Google search again in German, Dutch and English. No new DAs were found in this search in Dutch and English. We did find some updated versions of DAs in German (namely: DA2, DA4 and DA5) but after careful comparison we concluded that no changes were made that impacted the scores of the original DAs so we decided not to replace them. We also ran our academic search in PubMed once more for the period of April 2019-February 2021. This search identified 378 articles, but after title and abstract screening, none of the articles were selected for inclusion. Reasons for exclusion were: "no DA discussed", "not a treatment DA for CRC stage I-III" (e.g., a DA about screening decisions or metastatic cancer), and "not a DA but a nomogram."*

6. **Appendix IV: The reader should be able to understand the table in isolation but this table does not have a caption. For example, it might be better to change the heading for the second column to "Decision Aids (DA)" or provide a caption (or footnote) explaining all the initialisms.**

Agreed, we have now provided the following caption:

*"**Table 5:** The Kappa statistic per team. Explanation of the cells and titles: 'Reviewer team' consists of two reviewers (indicated by the initials of the reviewer (SH, RV, FC, MB, EK or XV)); 'DAs' = Decision Aids (N = number of DAs reviewed by this ream (DA numbers that are reviewed by this team)); 'Kappa IPDAS' = the kappa statistic relating to the International Patient Decision Aids Standards (rounded kappa (unrounded kappa)); 'Kappa CA' = the kappa statistic related to the Communicative Aspects checklist (rounded kappa (unrounded kappa))."*

7. **Appendix V: The first table is named "Table 1" but that name has already been used in the manuscript. Please, consult the author guidance on naming of tables within appendices.**

Thank you for spotting this, we have numbered the tables correctly now (starting from Table 5 in the Appendix) to avoid confusion.

8. **A non-essential suggestion: Provide the agreement matrices as a separate appendix in a file type more suitable for subsequent analysis, e.g. CSV, TXT (both for easy import), XLSX, RDS (for R), MAT (for MATLAB).**

Unfortunately, all appendices have to be uploaded in PDF so this is not possible (however, we have kept the lay-out as simple as possible so readers can easily copy and paste the numbers into their preferred format).

Reviewer 2:

1. **Thank you for the giving me the opportunity to read this manuscript. It was well written. There are some noticeable typing errors (e.g. the spelling of the word "Strengts" on page 2 line 50 and page 4 line 4 the word " clinical pracitce." to point out a couple of errors, please do check the document full for similar errors.**

Thank you for spotting these errors, we have changed them. We have not found any other typing errors.

2. **The final papers used were 18, any reason the region does not include Asian countries that publish in English or even any paper from England?**

Thank you for pointing that out. That is a coincidence. There were no exclusion criteria based on region, but we simply did not find any decision aids that were in English (or German or Dutch) but published in the Asian region.

Acknowledging that decision aids from other regions could be distinctly different, we included the following section in our discussion (also based on comments made by reviewer 1 in the previous round of revisions):

"*Although it has been demonstrated that culture might impact the effectiveness of health communication between doctors and patients of different cultural background*[39-40]*, studying cultural differences between European countries remains challenging as theories and methods for assessing differences vary between countries*[41]*. Future reviews could look into systematic differences between DAs from different countries more to see if shared decision-making is a globally agreed upon goal.*"